

1. Administrative & Study Identification

Full Study Title: A Phase 1 Open-label Study to Evaluate the Safety and Efficacy of MK-8294 Monotherapy in Advanced Solid Tumors **Short Title/Acronym:** MK-8294-001 **Protocol Number:** MK-8294-001 **NCT Number:** NCT07030712 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** MK-8294 (Nectin-4 T-cell engager / DAB014236) **Phase of Development:** Phase 1 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of MK-8294 monotherapy and to find the highest dose level (Maximum Tolerated Dose) that participants can tolerate. **Secondary Objectives:** To evaluate the anti-tumor activity of MK-8294, including the Objective Response Rate (ORR).

2.2 Background & Rationale To address the unmet medical need in patients with advanced or metastatic solid tumors (including head and neck squamous cell carcinoma, cervical squamous cell carcinoma, esophageal squamous cell carcinoma, triple-negative breast cancer, ER/PR+ HER2- breast cancer, endometrial, and bladder cancer). The study provides a new targeted immunotherapy option for those who have previously failed standard treatment, lack standard treatment options, or are intolerant to standard therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose escalation) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 67 participants **Number of Sites:** Global multicenter (including sites in the US, Netherlands, and Israel) **Estimated Study Start:** August 2025 **Estimated Primary Completion:** Up to 2 years (ongoing observation until disease progression or toxicity)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed advanced/metastatic solid tumor (e.g., head and neck, cervical, esophageal, breast, endometrial, or

bladder cancer) with measurable disease per RECIST 1.1. 3. Have previously failed standard treatment, lack standard treatment options, or are intolerant to standard treatment.

4.2 Exclusion Criteria 1. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 2. Active autoimmune disease that has required systemic treatment in the past 2 years (except for hormone replacement therapies). 3. History of New York Heart Association (NYHA) Class II or greater heart failure, or history of stem cell/solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Escalating doses starting from 30 µg up to a planned 70 mg. Administered on Day 1, Day 8, and Day 15 of each cycle (each cycle = 21 days). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** No maximum number of cycles; continuous 21-day cycles until disease progression, unacceptable toxicity, or discontinuation criteria are met.

5.2 Concomitant & Prohibited Therapy Permitted: Replacement therapies (e.g., thyroxine, insulin, or physiologic corticosteroid), Antiretroviral Therapy (ART) for well-controlled HIV, and antiviral therapy for HBV/HCV. **Prohibited:** Systemic corticosteroids for radiation-related toxicities, treatment for an additional progressing malignancy, and systemic therapies for active infections.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging (RECIST baseline), Safety Labs
Baseline	Day 0	Enrollment, First Dose, Vital Signs
Interim	Days 1, 8, 15 (Q21D)	IV Infusion of MK-8294, Safety Labs, Efficacy Assessment, optional CD8 PET imaging
End of Study	At progression/toxicity	Final Vital Signs, Exit Interview, AE Monitoring, Off-treatment follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Number of participants who discontinue study treatment due to an Adverse Event (AE), characterizing Dose-Limiting Toxicities (DLTs). **Measurement Method:** Clinical safety assessments, laboratory findings, and AE recording at each visit.

7.2 Secondary & Exploratory Endpoints Objective Response Rate (ORR): Percentage of participants with a Complete Response (CR)

or Partial Response (PR) evaluated per RECIST Version 1.1.
Exploratory: Evaluation of CD8 PET tracer distribution in optional imaging subsets.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Untoward medical occurrences temporally associated with the study treatment will be recorded continuously. Immune-related AEs will be closely monitored.

Serious Adverse Events (SAEs): Reported within 24 hours to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** Internal safety review committee governing the transition and authorization between dose-escalation cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving ≥ 1 dose of MK-8294); Efficacy Set (participants with measurable disease at baseline per RECIST 1.1). **Sample Size Justification:** $N = 67$ participants, statistically mapped for traditional Phase 1 dose-escalation algorithms to determine the MTD. **Handling of Missing Data:** Missing AE data will not be imputed; tumor response endpoints will utilize established oncological handling methodologies (e.g., LOCF or censoring per RECIST guidelines).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring to verify source data and protocol compliance. **Audit Trail:** eCRF locking, structured cohort safety reviews, and data clarification forms tracking via the trial's validated electronic data capture (EDC) system.

11. Study Identifiers & Governance

Primary ID: MK-8294-001 **Registry Number:** NCT07030712 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Clinical Study of MK-8294 in Participants With Advanced Solid Tumors (MK-8294-001) **Official Regulatory Title:** A Phase 1 Open-label Study to Evaluate the Safety and Efficacy of MK-8294 Monotherapy in Advanced Solid Tumors

12. Clinical Framework (Oncology Specific)

(Note: Adapted for Oncology framework) **Primary Condition:** Advanced/Metastatic Solid Tumors **Diagnosis Threshold:** Histologically or cytologically confirmed metastatic disease, failure on standard therapies **Medical Methodology:** Targeted immunotherapy (Nectin-4 T-cell engager) **Optimization Target:** Maximum Tolerated Dose (MTD) determination

13. Study Procedures & Methodology

Management Pathway: Phase 1 Oncology Dose-Escalation Pathway **Education Protocol (HCP):** Investigator Brochure review and MK-8294 clinical protocol training **Education Protocol (Patient):** Informed consent outlining trial expectations, infusion schedules, and AE reporting guidelines **Quality Audit Mechanism:** Clinical safety data reviews before escalating to the next dosing cohort

14. Observation & Follow-Up Schedule

Total Duration: Up to ~2 years (continuous cycles until progression/unacceptable toxicity) **Onsite Visit Cadence:** Days 1, 8, and 15 of every 21-day cycle **Remote Monitoring Frequency:** Inter-cycle monitoring dependent on clinical status and AE follow-up **Checklist Review Interval:** Prior to every 21-day cycle administration

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Insert Name]	_____	03-Apr-2026				
Clinical Operations Lead	[Insert Name]	_____	03-Apr-2026				
Lead Statistician	[Insert Name]	_____	03-Apr-2026				